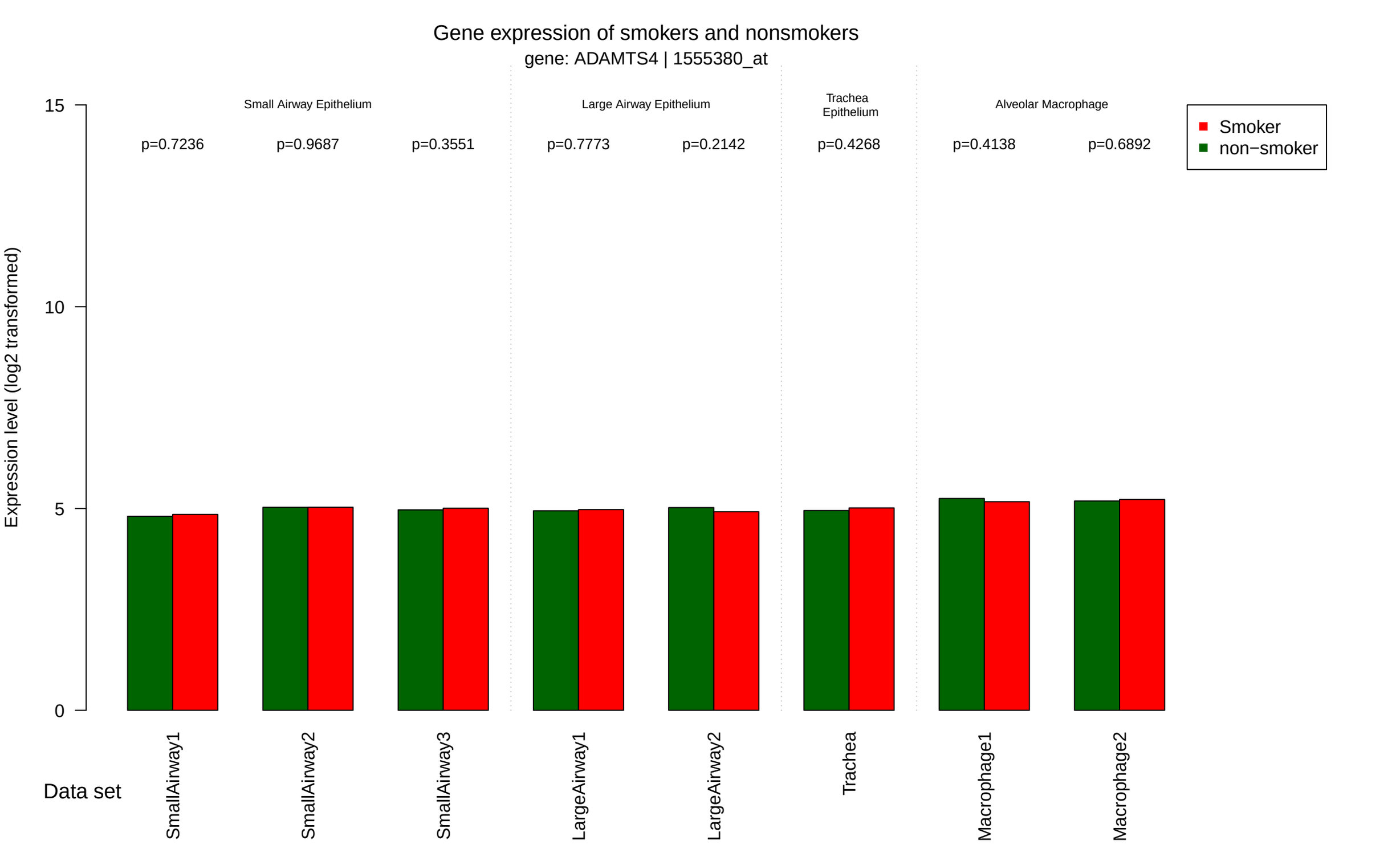
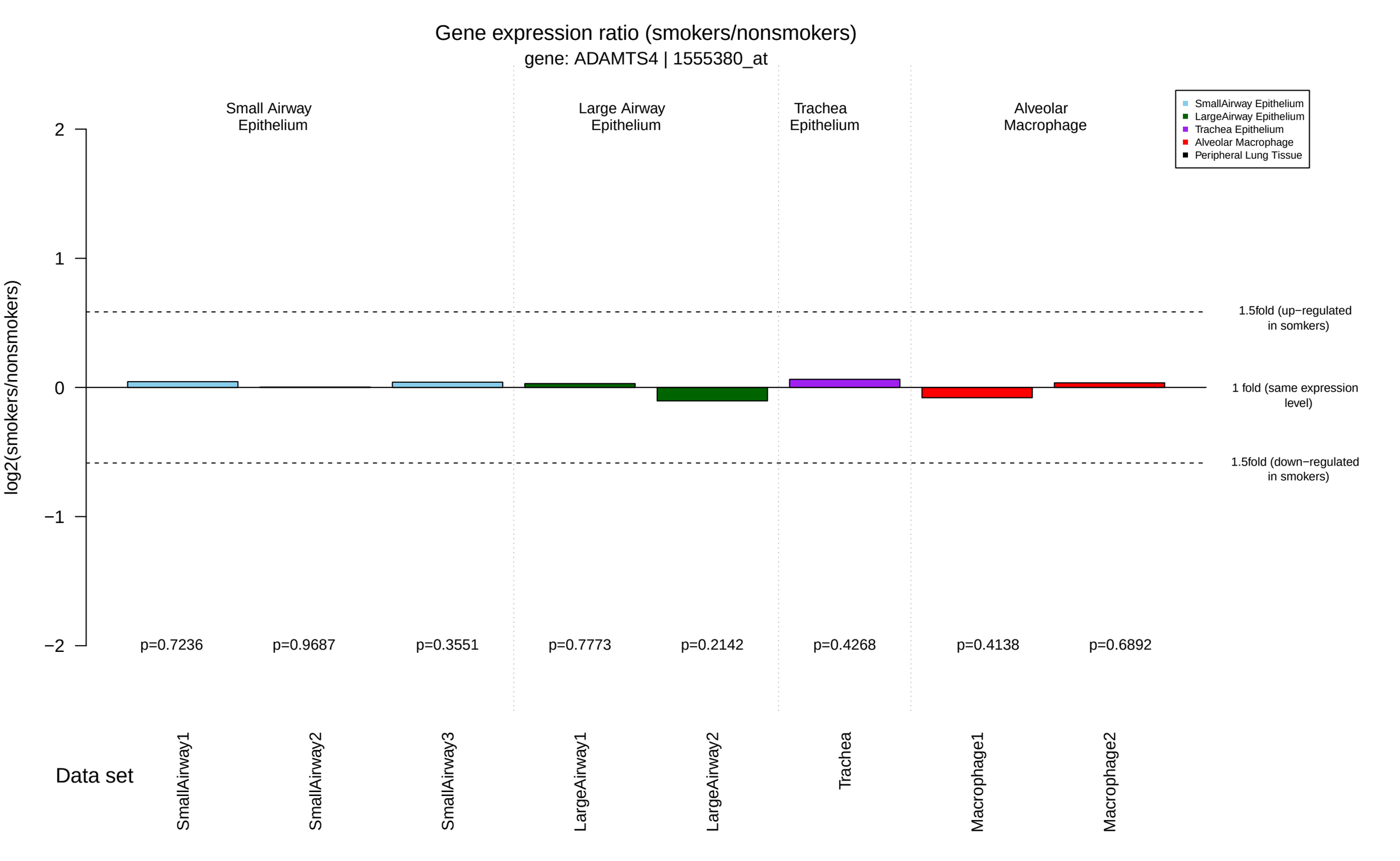
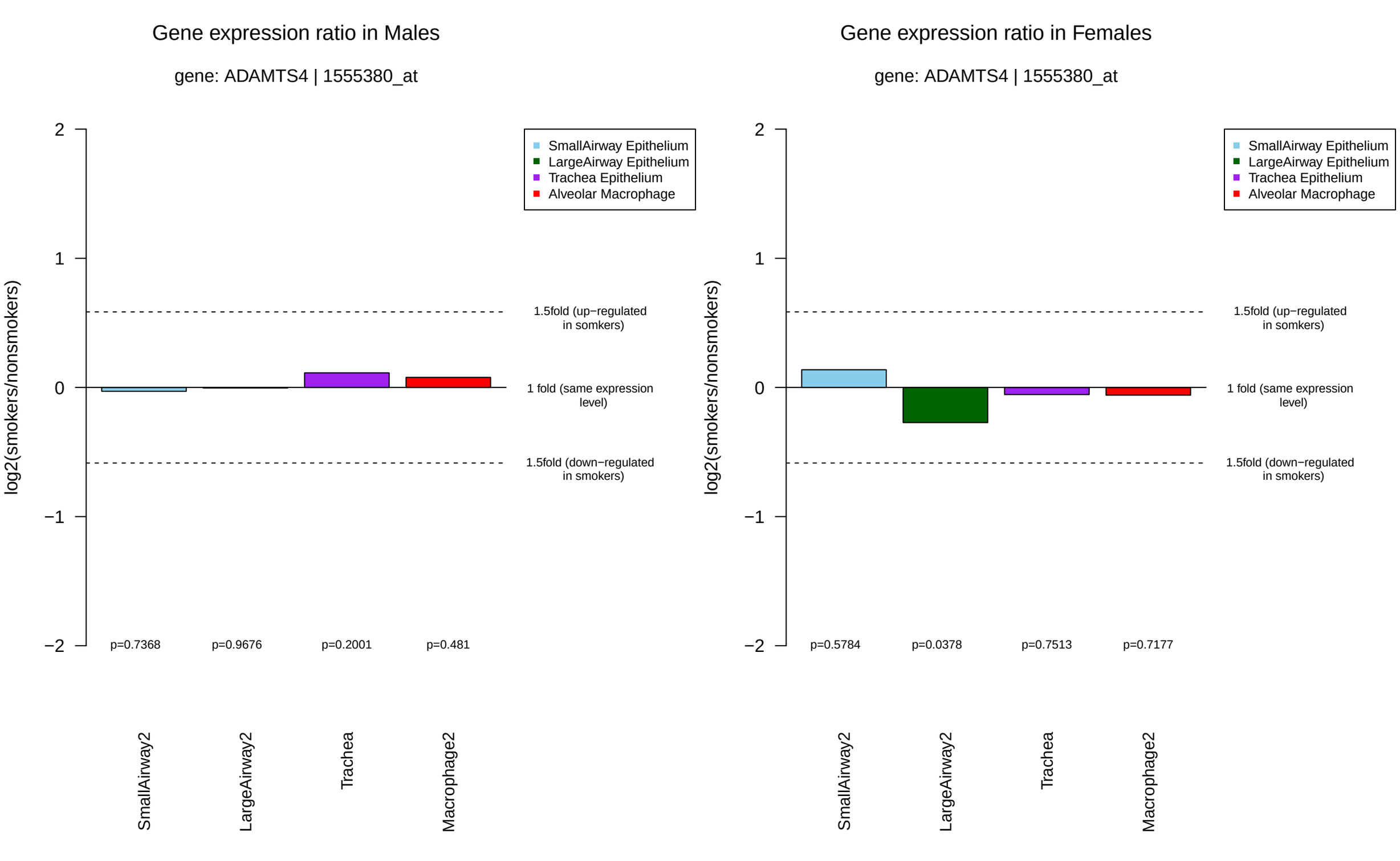
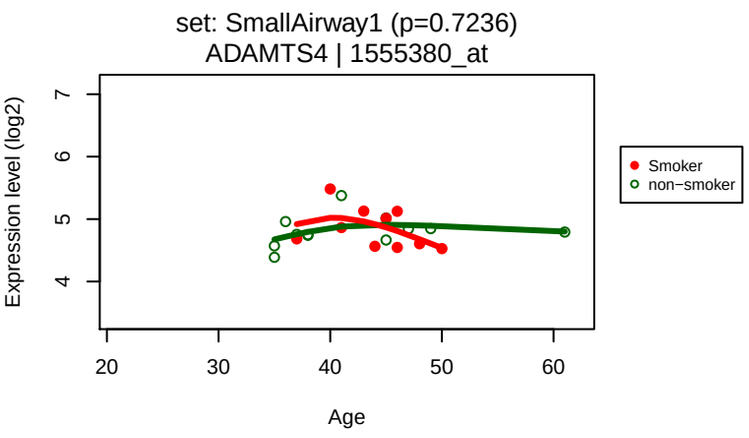
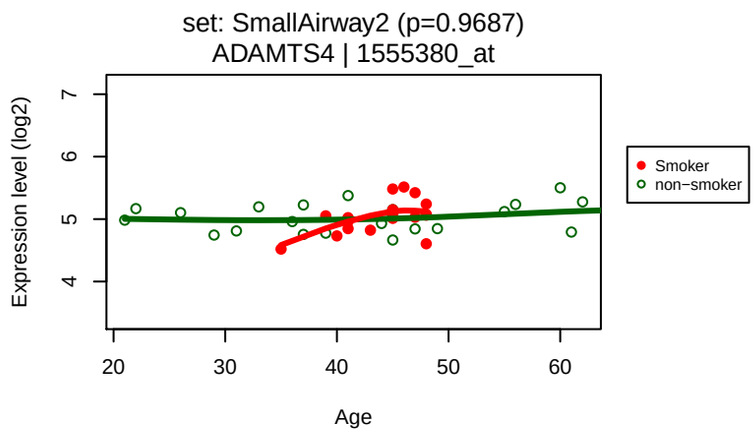
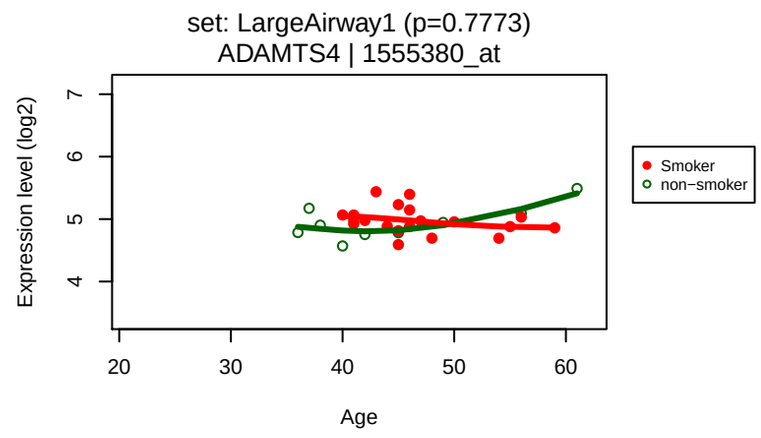

Gene expression of smokers and nonsmokers
gene: ADAMTS4 | 1555380_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7236
p=0.9687
p=0.3551
p=0.7773
p=0.2142
p=0.4268
p=0.4138
p=0.6892
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

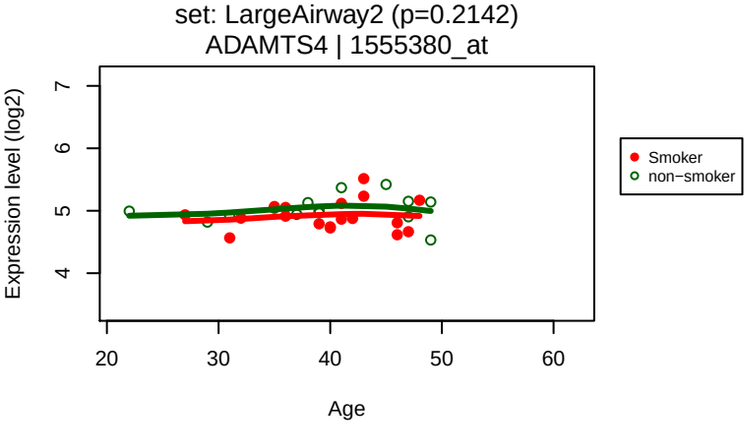

Gene expression ratio (smokers/nonsmokers)
gene: ADAMTS4 | 1555380_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7236
p=0.9687
p=0.3551
p=0.7773
p=0.2142
p=0.4268
p=0.4138
p=0.6892
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

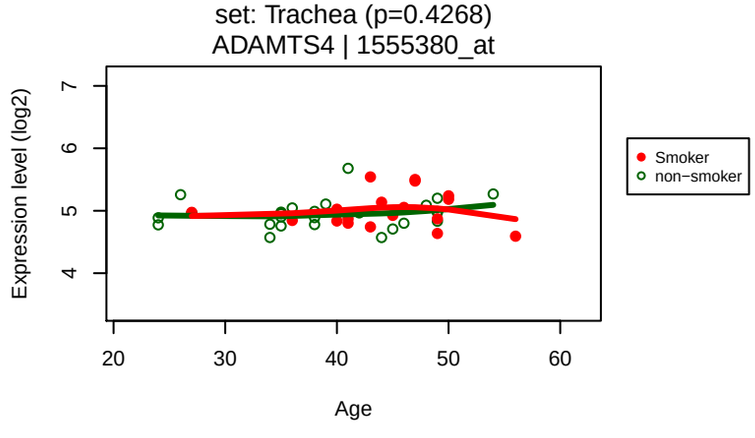

Gene expression ratio in Males
gene: ADAMTS4 | 1555380_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7368
p=0.9676
p=0.2001
p=0.481
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADAMTS4 | 1555380_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5784
p=0.0378
p=0.7513
p=0.7177
SmallAirway2
LargeAirway2
Trachea
Macrophage2

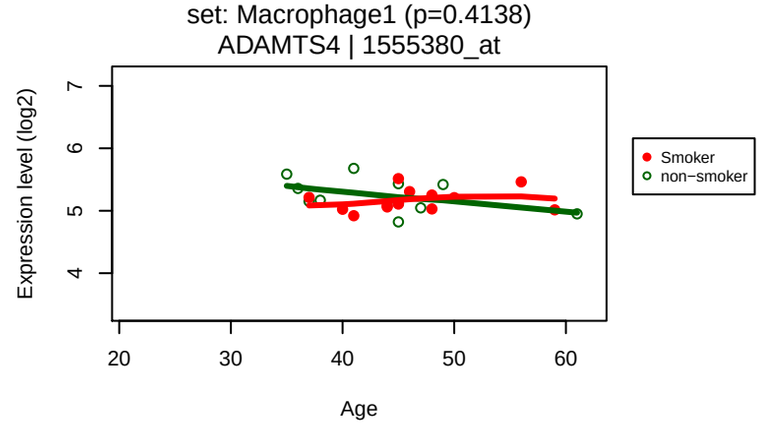

set: SmallAirway1 (p=0.7236)
ADAMTS4 | 1555380_at
Expression level (log2)
Age
Smoker
non−smoker

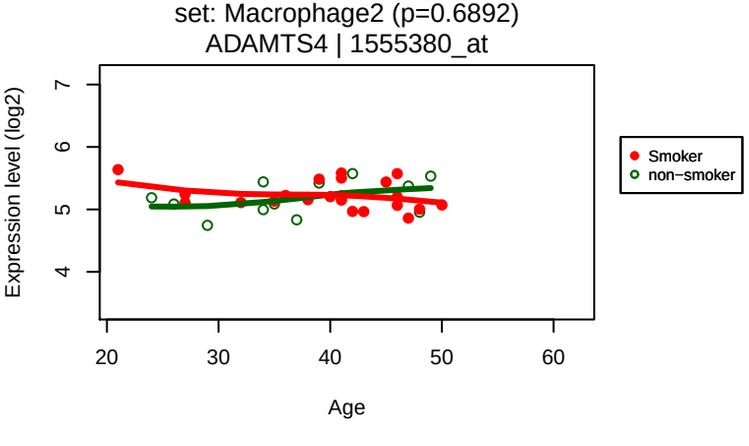

set: SmallAirway2 (p=0.9687)
ADAMTS4 | 1555380_at
Expression level (log2)
Age
Smoker
non−smoker

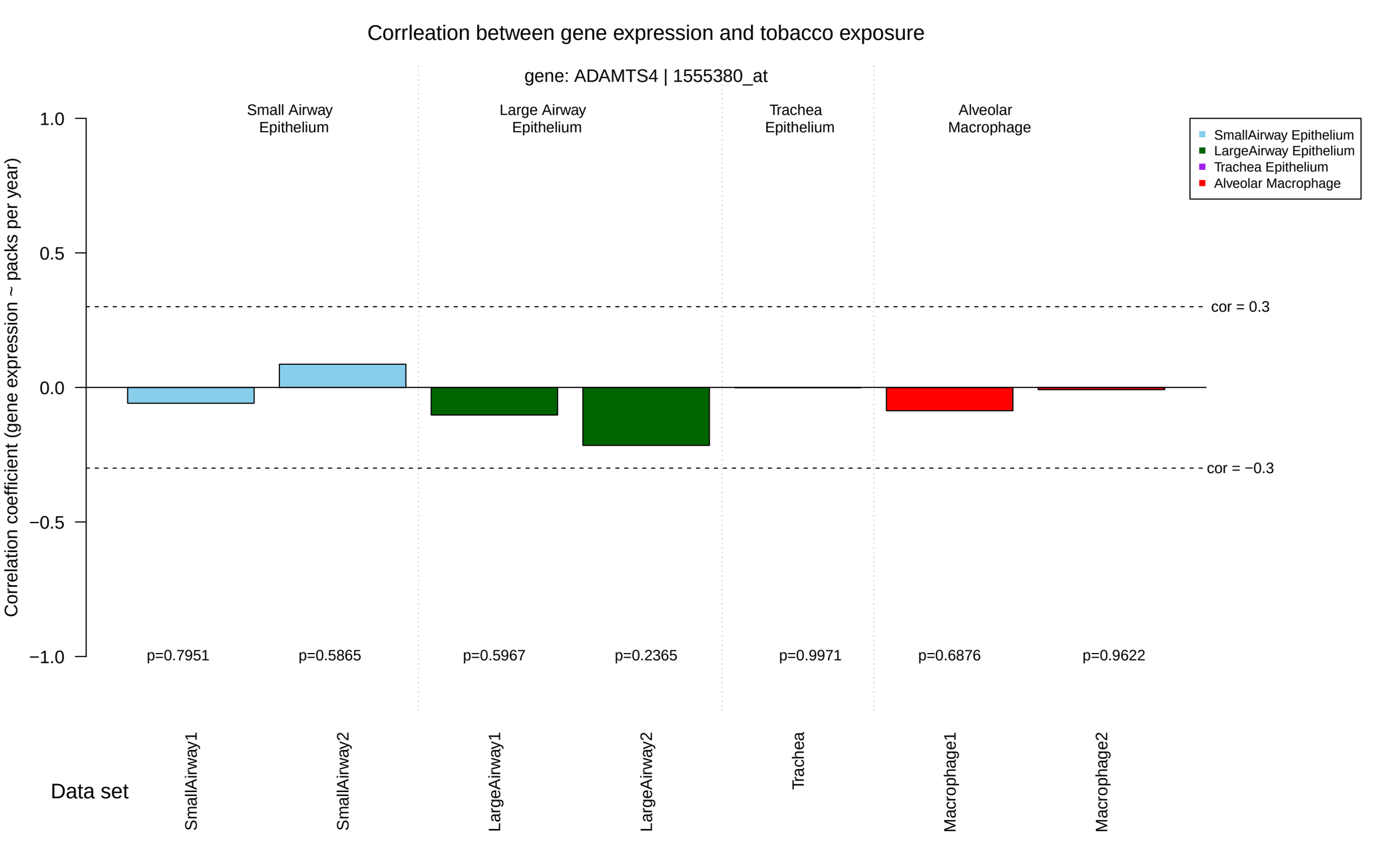

set: LargeAirway1 (p=0.7773)
ADAMTS4 | 1555380_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2142)
ADAMTS4 | 1555380_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4268)
ADAMTS4 | 1555380_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4138)
ADAMTS4 | 1555380_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6892)
ADAMTS4 | 1555380_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADAMTS4 | 1555380_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.7951
p=0.5865
p=0.5967
p=0.2365
p=0.9971
p=0.6876
p=0.9622
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2